PROJECT REPORT

Title: ClaimVision — AI-Powered Vehicle Insurance Cost Estimation Platform

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1. INTRODUCTION

Vehicle insurance claim processing is traditionally slow, manual, and prone to fraud. It involves multiple stakeholders, paperwork, and physical inspections, which leads to delays and inefficiencies.

ClaimVision is an AI-powered platform that automates the entire insurance claim lifecycle. It allows users to upload vehicle damage images and insurance documents, after which AI models analyze the damage and estimate repair costs. The system then enables officers and administrators to review, approve, and process claims digitally.

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2. OBJECTIVES

- To automate vehicle damage detection using AI
- To estimate repair costs accurately using machine learning
- To reduce claim processing time
- To minimize fraudulent claims
- To provide a complete digital claim-to-payout workflow
- To improve transparency between users and insurance companies

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3. SYSTEM OVERVIEW

ClaimVision is an end-to-end platform consisting of multiple modules:

- User uploads images and documents
- AI engine analyzes damage and predicts cost
- Officers review and verify claims
- Admin provides final approval
- Cashier initiates payout

Workflow:

User → AI Analysis → Officer Review → Admin Approval → Cashier → Payout

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4. SYSTEM ARCHITECTURE

The system is composed of the following components:

- Frontend: User interface for interaction
- Backend: API and business logic
- ML Engine: AI models for analysis

- Database: Stores structured data

- Cloud Storage: Stores images and documents

Architecture Flow:

[User Upload]

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[AI Engine]

- Damage Detection
- Heatmap Generation
- Cost Prediction

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[Officer Portal]

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[Admin Portal]

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[Cashier System]

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[Payout System]

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5. AI / ML ENGINE

5.1 Damage Detection Model

- Input: Vehicle images (front, rear, left, right)

- Model: YOLOv8 / Detectron2

- Output: Bounding boxes and damage classification

Damage Types:

Dent, Scratch, Crack, Shatter, Deformation, Missing Parts

5.2 Heatmap Generation

- Technique: Grad-CAM

- Purpose: Visualize damage severity

Color Representation:

Red → Severe damage

Yellow → Moderate damage

Green → Minor damage

5.3 Cost Prediction Model

- Model: XGBoost / LightGBM

Input Features:

- Vehicle company and model

- Manufacturing year

- Damage type and severity

- Damaged parts

- Insurance company

- Location (labour cost variation)

- User claim history

Output:

Estimated repair cost range (minimum to maximum in INR)

5.4 Document OCR

- Tools: Tesseract OCR / AWS Textract

Extracted Details:

- Policy number

- Coverage type

- Expiry date

- Insured name

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6. USER MODULE

Users can:

- Register and log in
- Submit insurance claims
- Upload vehicle images and documents
- View AI-generated damage analysis
- Track claim status
- Receive payout

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7. CLAIM FORM DETAILS

Personal Information:

- Full Name
- Date of Birth
- Aadhaar / PAN
- Mobile Number
- Email ID
- Address

Vehicle Information:

- Registration Number
- Manufacturer and Model
- Year of Manufacture
- Fuel Type

- Engine and Chassis Number

Insurance Details:

- Policy Number
- Insurance Company
- Policy Type
- Validity Period

Incident Details:

- Date and Time
- Location
- Type of Incident
- FIR Details (if applicable)

Damage Details:

- Uploaded images
- AI-detected damage
- Heatmap visualization

Bank Details:

- Account Number
- IFSC Code
- UPI ID

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8. ROLE-BASED ACCESS SYSTEM

User:

- Submit and track claims

Field Officer:

- Review claims and approve/reject

Senior Officer:

- Override decisions

Admin:

- Full system control

Insurance Company:

- Monitor their claims

Cashier:

- Process payouts

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9. CLAIM PROCESSING WORKFLOW

1. User submits claim

2. AI analyzes images and estimates cost

3. Claim marked as "Under Review"

4. Field Officer reviews claim

5. Senior/Admin approval

6. Cashier processes payment

7. User receives payout

Final Output Includes:

- Approved amount

- Digital bill

- QR verification

- Digital signature

- Timestamp and claim ID

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10. TECHNOLOGY STACK

Frontend:

- Next.js

- Tailwind CSS

- React Hook Form

Backend:

- Node.js / Express or FastAPI

- PostgreSQL

- Redis

Machine Learning:

- Python
- YOLOv8
- XGBoost
- Grad-CAM

Storage:

- AWS S3 / Cloudinary

Payments:

- Razorpay

DevOps:

- Docker
- GitHub Actions
- AWS / Vercel

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11. SECURITY FEATURES

- OTP-based authentication
- AES-256 encryption for documents
- JWT-based authorization
- Audit logs

- Image tampering detection
- Secure payout with PIN verification

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12. FUTURE ENHANCEMENTS

- 3D vehicle damage visualization
- Real-time chat system
- Fraud detection using AI
- Multi-language support (Tamil, Hindi, English)
- WhatsApp integration
- Mobile application
- Blockchain-based bill verification

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13. PROJECT SIGNIFICANCE

- Solves a real-world insurance problem
- Combines multiple AI technologies
- Provides complete system implementation
- Includes real payment integration
- Supports explainable AI
- Has research and industrial relevance

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14. CONCLUSION

ClaimVision provides an efficient, intelligent, and secure solution for vehicle insurance claim processing. By integrating AI-based damage detection, cost prediction, and a structured approval workflow, the platform significantly improves speed, accuracy, and transparency.

This project demonstrates the practical application of modern technologies in solving real-world challenges.
